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1 Introduction

This section first explains what *engineering systems* are, why their long lifetimes and irreversibility make planning particularly challenging, and how planning is structured at different levels. We then contrast the conventional strategic planning process with a more flexible approach, highlighting the advantages of flexibility in addressing uncertainty and adapting to future developments.

Learning Objectives

After working through Section 1,

- you can **explain** the terms *engineering system* and *flexibility*;
- you can **describe** both the conventional and the flexible strategic planning process for engineering systems; and
- you can **explain** why the conventional planning process is criticized in the literature and how the flexible planning process intends to address these deficits.

1.1 Engineering Systems

The term *engineering system* refers "complex systems in the aerospace, defense, energy, housing, telecommunication, and transportation industries" (Cardin, 2014, p. 2). Examples of engineering systems include power plants, railway stations, airports, office complexes, or bridges.

A common characteristic of engineering systems is their long lifetime, often exceeding 20 years. Over such long time spans, future developments are highly uncertain. To illustrate this, consider the case of a motorway bridge over a large river. While the bridge is designed to meet current demand, it is unclear whether traffic volumes in the region will increase, decrease, or even disappear entirely in 20, 50, or 100 years. Such uncertainty makes long-term planning particularly challenging.

In addition to uncertainty, engineering systems usually require substantial upfront investments that are largely *irreversible*. Once built or installed, there is generally no secondary market in which such assets can be sold at a profit. Returning to the example of the motorway bridge: if the bridge is no longer needed in 50 years, there is no realistic way to dismantle and resell it for profit. For this reason, economists emphasize that engineering systems often have little or no *salvage value*.

Because of their durability and long lead times, engineering systems must be planned well in advance. Planning in this context is typically distinguished by the literature across three levels (De Neufville et al., 2013; S. Shuchi et al., 2012; S. N. Shuchi, 2016):

- **Operational planning** deals with day-to-day activities, such as staffing or stakeholder agreements. Given its short time horizon, uncertainty is relatively limited, though not absent.
- **Tactical planning** covers a time range of roughly two to five years. It focuses on how the system can be adapted and improved within its existing perimeter, often with the aim of optimizing processes or components.
- **Strategic planning** is concerned with long-term visions for the system. It defines how the system should be adapted, expanded, or rebuilt to meet future needs. Because of

its extended planning horizon, strategic planning is subject to particularly high levels of uncertainty.

In what follows, we will focus exclusively on the strategic planning of engineering systems. Operational and tactical planning will not be discussed further. More specifically, we will examine the so-called conventional strategic planning process, which has long been regarded as the de facto standard for long-term infrastructure planning. We then contrast it with a more flexible strategic planning process that aims to overcome the limitations of the conventional approach.

1.2 Conventional Strategic Planning

Figure 1 shows a generic conventional strategic planning process, which consists of five main steps:

Step 1: *Inventory:* In the first step, an inventory of the existing system is created. Here, planners record the condition of the system, such as available building areas, structural characteristics, or throughput capacity. For greenfield projects, the inventory focuses on local conditions such as available space or soil quality.

Step 2: *Forecast:* In the second step, a forecast is developed to describe the expected future of the system. Various methods are used in practice: for example, the *Delphi method*, in which scenarios are generated based on expert opinion, or extrapolation methods such as regression models. In conventional strategic planning, planners usually determine the future scenario, which they deem most probable.

Step 3: *Facility Requirements:* In the third step, facility requirements are defined. These requirements specify when, how, and where capacity needs to be updated, expanded, or modified, depending on the forecast.

Step 4: *Alternatives:* In the fourth step, planners identify and develop alternatives that show how these facility requirements can be implemented in practice.

Step 5: *Master Plan:* Finally, in the fifth step, one alternative is selected and documented in a master plan or program, which sets out the long-term vision for the system.

Although this process appears straightforward, the literature mentions that it has important shortcomings (De Neufville & Scholtes, 2011). One major limitation is that the forecast in step two usually consists of only one or a few scenarios describing the "most probable" future. In practice, this approach is often highly inaccurate because strategic planning horizons extend over 20–50 years, during which future developments cannot be known with certainty. In addition to the limited treatment of uncertainty, the literature criticizes other aspects of the conventional process. Facility requirements are typically defined as fixed and static, ignoring the ability of managers and planners to adapt to changing conditions. The master plan, once written, often assumes that decisions must be adhered to for decades, regardless of how circumstances evolve (De Neufville et al., 2013).

1.3 Flexible Strategic Planning

The flexible planning process, which is illustrated in Figure 1, addresses the shortcomings of conventional strategic planning by treating plans not as rigid roadmaps but as adaptive frameworks. A flexible engineering system is defined in the literature as one that can change easily in response to uncertainty (De Neufville & Scholtes, 2011; Magalhães et al., 2013; Saleh et al., 2002), for example by altering its mode of operation or technical attributes

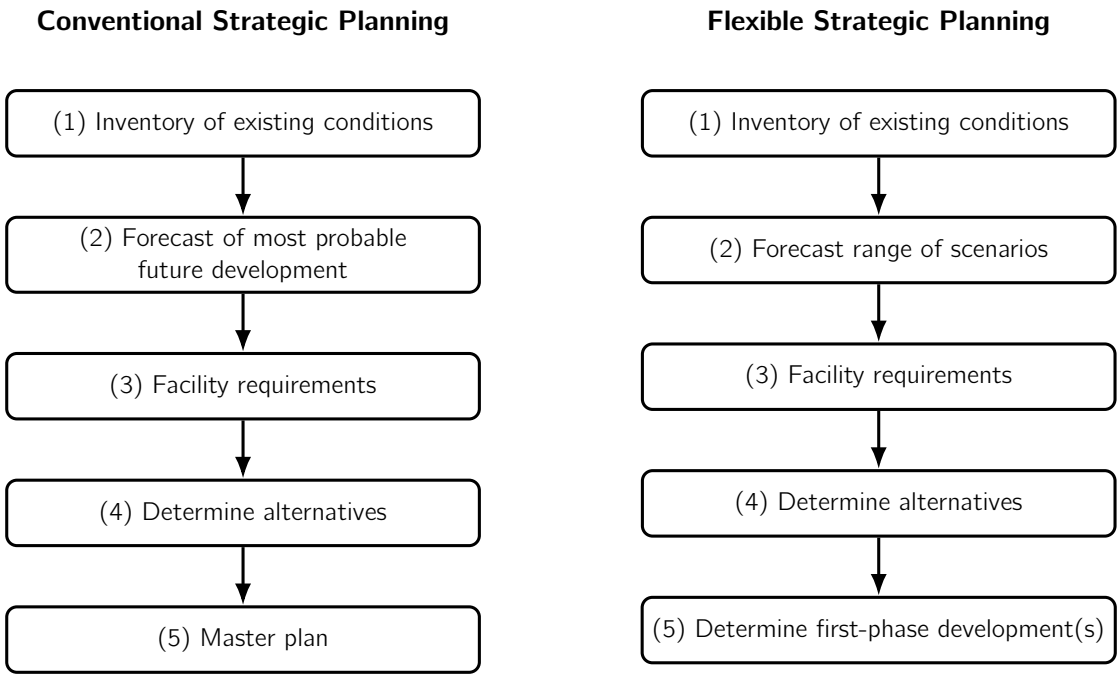

Figure 1: Comparison of a conventional planning process (left) and a flexible planning process (right)

(Wang & De Neufville, 2005). By being able to adapt, flexible systems can mitigate the effects of negative developments and capitalize on positive ones. For this reason, they are often more valuable than conventional systems: empirical studies suggest value gains of 10–30 % (Cardin, 2014).

The flexible strategic planning process differs from the conventional one in three main steps. In step two, planners no longer generate a single "most probable" forecast but instead develop a wide range of possible future scenarios (Indeed, thousands of different scenarios are often created). In step three, facility requirements are then derived in a way that is robust across the entire range of scenarios, rather than tailored to a single forecast. Finally, in step five, planners do not produce a fixed master plan. Instead, they define a *first-phase development* that specifies only the initial steps. All subsequent decisions are deliberately postponed to the future, when more information will be available.

For the remainder of this document, we will focus on selected aspects of flexible strategic planning. Section 2 discusses methods to generate forecasts that explicitly incorporate uncertainty. Section 3 introduces the concept of real options and shows how flexibility can be embedded in engineering systems. Section 4 explains how to formulate strategic plans for a conventional engineering system, while Section 5 focusses on flexibly planned engineering systems. Finally, Section 6 explains how planner can select the optimal strategic plan for an engineering system.

2 Recognition of Uncertainty and Modelling Scenarios

This section introduces the role of uncertainty in long-term planning for engineering systems and motivates why single "most-likely" forecasts are insufficient. In a first step, we consider a real-world planning example considering Madrid Barajas Airport, which illustrates perfectly that forecasts are subject to uncertainty. Then, we define types and sources of uncertainty.

Finally, we outline and apply a practical five-step framework for recognizing, estimating, and modeling uncertainty to generate scenario-based forecasts.

Learning Objectives

After working through Section 2,

- you can **explain** the term *uncertainty*, what types of uncertainty there are, and what their origin is;
- you can **describe** the 5-stepped process used to recognise and model uncertainty; and
- you can **apply** the 5-stepped process used to recognise and model uncertainty to a (simple) practical example.

2.1 The Forecast is Always Wrong

In 1999, the Spanish *Ministry of Public Works and Transportation* approved the master plan for Madrid Barajas Airport, the country's largest airport. The plan, covering the period 1999–2025, was based on a single demand forecast, illustrated in Figure 2.

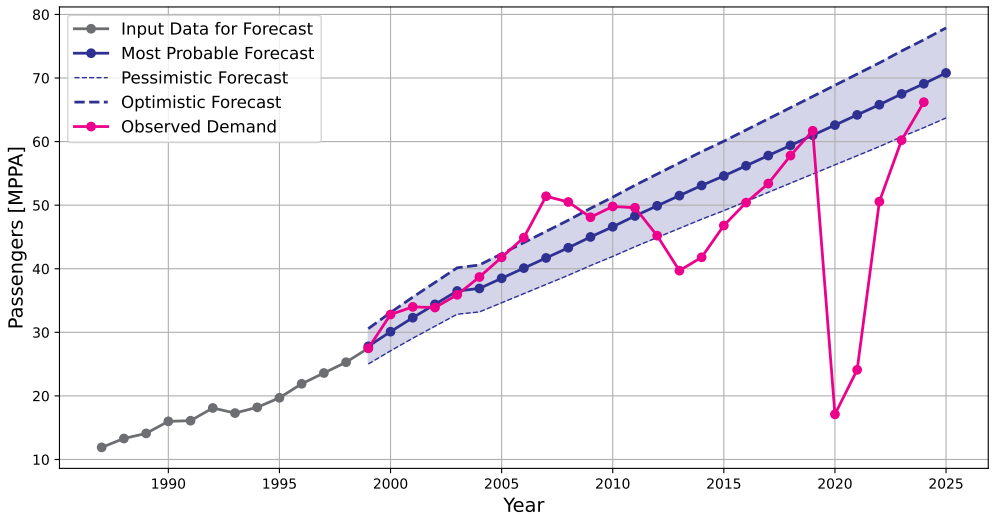

Figure 2: Observed Demand vs. Demand Forecast for Madrid Barajas Airport. Input data from AENA (2025).

The forecast included a baseline scenario (solid blue line), considered by the planners to be the most probable future, as well as optimistic and pessimistic variants (dashed blue lines). According to these projections, demand at Barajas Airport was expected to rise from about 27 million annual passengers in 1999 to roughly 70 million in 2025. On this basis, the airport management decided to substantially expand terminal capacity. In 2006, the expansion was completed at a total cost of 6.2 billion euros, increasing terminal space to more than one million square meters. At that time, Madrid Barajas Airport was home to the largest airport terminal complex in the world.

In retrospect, the 1999 forecast proved highly inaccurate. As shown in green in Figure 2, it underestimated actual demand between 1999 and 2011, only to overestimate it thereafter. The result of the 1999 Master Plan was that new large terminals were built for which sufficient demand never materialized.

The Madrid Barajas example illustrates a broader problem: with planning horizons of 20–50 years, demand forecasts for engineering systems are inevitably subject to substantial uncertainty. This uncertainty is so great that MIT professor *Richard de Neufville* famously remarked that **forecasts are always wrong** (De Neufville et al., 2013). Consequently, planners should rely on forecasting approaches that explicitly incorporate uncertainty rather than depending on a single "most probable" scenario. In the following, we will examine methods and tools that enable such an approach.

2.2 What is Uncertainty?

Uncertainty can be defined as "any departure from the unattainable ideal of complete determinism" (Walker et al., 2003, p. 4). Determinism implies that all events are fully determined by preceding causes. By contrast, uncertainty arises when not all events can be explained by observable causes but instead occur in ways that appear random.

In the literature, four different types of uncertainty are typically distinguished (De Weck et al., 2007; Walker et al., 2003):

- **Known uncertainty** can be described and quantified on the basis of historical observations.
- **Unknown uncertainty** refers to events that are entirely unforeseeable and fundamentally unpredictable. A prominent example is the COVID-19 pandemic and its wide-ranging effects on demand and sales figures.
- **Uncertainty in the data** arises during data collection, for example when measurements are inaccurate, incomplete, or missing.
- **Uncertainty in the description** results from incomplete, ambiguous, or inconsistent definitions of concepts or variables.

For the purposes of this course, we will focus primarily on *known uncertainties*. These can originate from different sources. The literature distinguishes between **endogenous uncertainty**, whose causes lie within the system, and **exogenous uncertainty**, which arises from outside the system.

- Endogenous uncertainty may be linked to the reliability of a product or to internal organizational decisions. For example, if a company that owns an engineering system changes its corporate strategy, the requirements imposed on that system may shift as a result.
- Exogenous uncertainty often arises from changes in the market, such as fluctuations in demand or the entry of competitors. Political and regulatory changes can also create significant uncertainty, as can shifts in the way a system is used—such as when users adopt it for purposes different from those originally intended.

2.3 Recognizing, Estimating, and Modelling Uncertainty

To address uncertainty in the planning of engineering systems, De Neufville and Scholtes (2011) propose a five-step process. This framework guides practitioners in identifying, analyzing, and modeling the factors most critical for the future performance of a system. The five steps, outlined below and discussed in more detail in the subsequent subsections, are as follows:

Step 1: In step one, important factors that are most crucial for the future performance of the system we are to design are identified.

Step 2: Subsequently, these factors are analyzed in a second step on the basis of historic data.

Step 3: In a third step, we examine whether these factors were subject to certain trend breakers in the past.

Step 4: In a fourth step, we look at how good forecasts for the identified factors have been in the past to find out whether we can trust the forecasts or not.

Step 5: And finally, in the fifth and last step, the identified factors of uncertainty are described by using a mathematical model.

In the following, we will go through this entire process step-by-step. For this purpose, we will focus on a practical example cornering on the capacity of the passenger building(s) at Zurich Airport.

2.3.1 Step 1: Identifying Important Factors and Drivers

To illustrate the five-step process, we begin with step one: the identification of the most important factors and drivers that influence the design of an engineering system.

When identifying factors, four aspects should be taken into account:

- *Type:* Numerous factors may influence an engineering system, including economic, technical, and regulatory aspects. The decisive factors are not always obvious, so it is often useful to think broadly.
- *Level of aggregation:* Once factors are identified, they must be specified precisely. For example, should we consider overall passenger demand, or focus on specific segments such as demand from low-cost carriers?
- *Variability over time:* Many factors fluctuate across different time scales. In the case of our Zurich Airport example, it is therefore important to determine whether it is sufficient to base strategic planning on annual aggregated demand, or whether short-term demand peaks must also be taken into account.
- *Usefulness:* Because the list of potential factors can be very long, experts must decide which are truly critical. In practice, it is often preferable to select a few key drivers rather than attempt to include too many.

Applying these considerations to Zurich Airport, several factors can be identified that affect terminal design: (i) passenger demand, (ii) demand expressed in terms of aircraft movements, (iii) available technology and its future development, and (iv) the average number of pieces of luggage carried by passengers. Since this is not a course in airport design, we will not examine all of these in detail. Instead, we select *passenger demand* as the key factor for further analysis.

In simplified terms, there is a direct relationship between annual aggregated passenger demand and the terminal space required: the higher the demand, the greater the space needed. Having identified this factor, we can now turn to step two, where we analyze historical demand data to better understand the uncertainty associated with it.

2.3.2 Step 2: Analyzing Historical Trends (and Data)

The second step of the process aims to understand the size, range, and form of uncertainty in the selected factor(s). De Neufville and Scholtes (2011) suggest three tasks for this step: (i) understand the data, (ii) identify general patterns, and (iii) assess the uncertainty. Again, we illustrate these tasks using passenger demand at Zurich Airport as an example.

1. *Understand the data:* Our dataset contains annual aggregated passenger demand at Zurich Airport from 1982 to 2022, shown as blue points in Figure 3. The data includes all departing, arriving, and transferring passengers per year. While these values allow us to analyze long-term changes, they do not reveal intra-annual variation. For instance, the dataset cannot tell us how passenger peaks in the summer of 1995 differed from those in 2010.

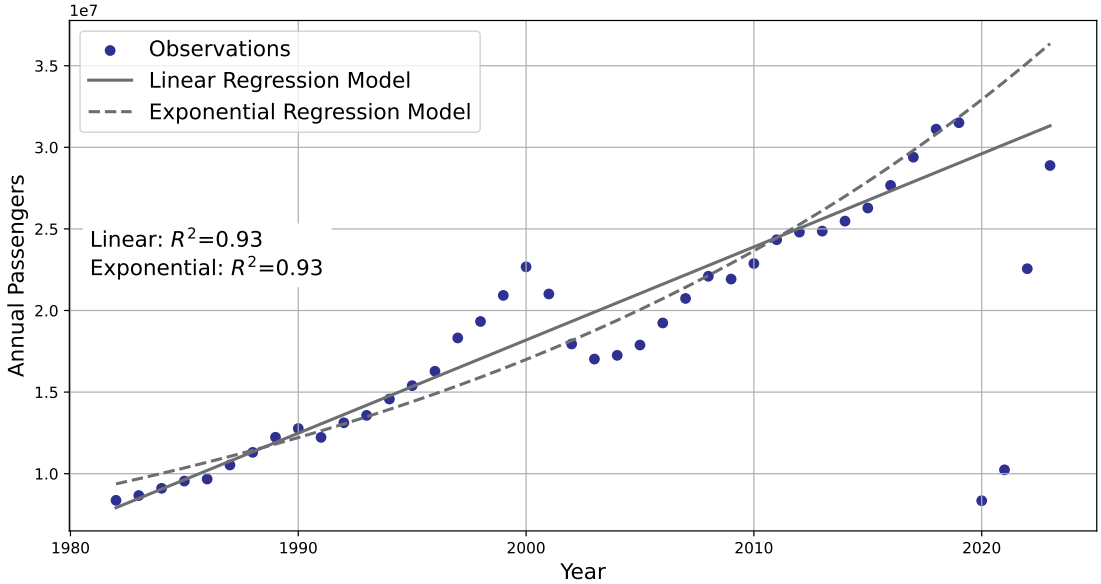

Figure 3: Annual Passenger Demand at Zurich Airport from 1982 to 2022

2. *Identify general patterns:* According to the literature, airport demand often grows exponentially (De Neufville et al., 2013; Horonjeff et al., 2010). To test this hypothesis, we performed regression analyses with both a linear function (solid black line in Figure 3) and an exponential function (dashed black line). In both cases, we excluded the data points of the years 2020 to 2022 from the regression analysis, as demand in these years was strongly affected by the COVID-19 pandemics. The resulting correlation coefficients are very high: 0.93 for the linear fit and 0.94 for the exponential fit.

3. *Assess uncertainty in the data:* After having a feeling for the data, we can now focus on how the data changes over the course of a year in relative terms. To quantify year-to-year variation, we calculate percentage growth using

$$\frac{\Delta D_t}{D_t} = \frac{D_{t+1} - D_t}{D_t}. \tag{1}$$

Our data set of aggregated demand observations for Zurich Airport covers the years 1982 to 2022, which corresponds to 38 data points ($t = 0, 1, 2, \ldots 37$). For all these data points, except the last data point for the year 2022 ($t = 37$), we can use

Equation 1 to describe the percentage growth for year t. For example, to calculate the percentage growth of demand in year 1983, we need to know the aggregated demand values for years $t = 1$ (i.e., year 1983) and $t = 2$ (i.e., year 1984). Applying Equation 1 to the full dataset yields the histograms in Figure 4, which display the distribution of percentage growth of demand at Zurich Airport with (left) and without (right) considering outliers. In this context, outliers correspond mainly to the COVID-19 years and will be discussed further in Step 3, see Section 2.3.4.

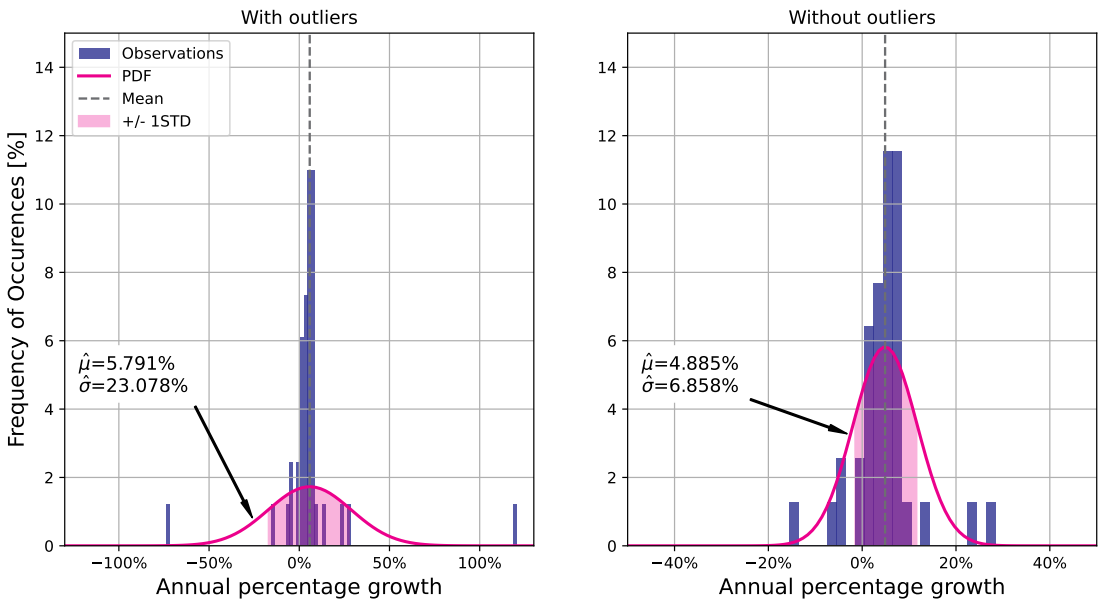

Figure 4: Percentage Change of Annual Air Traffic Movements (left) and Passengers (right) at Zurich Airport

To characterize the percentage growth data statistically, we can assume that the data is normally distributed. Therefore, we can fit a normal distribution with mean μ and standard deviation σ to the observed growth values. To this end, the mean percentage change $\hat{\mu}$ can be estimated[1] as

$$\hat{\mu} = \frac{1}{T} \sum_{t=0}^{T-1} \frac{\Delta D_t}{D_t} \quad \text{with } \Delta D_t = D_{t+1} - D_t \tag{2}$$

while the standard deviation $\hat{\sigma}$ can be estimated[2] as

$$\hat{\sigma} = \sqrt{\frac{\sum_{t=0}^{T-1} (\frac{\Delta D_t}{D_t} - \hat{\mu})^2}{T}}. \tag{3}$$

The fitted distributions are shown in red in Figure 4. For the data without outliers, the estimated mean annual growth rate is $\hat{\mu} = 4.885\,\%$ and the standard deviation is $\hat{\sigma} = 6.858\,\%$. For the data with outliers, it is especially evident that the estimated standard deviation of $\hat{\sigma} = 23.078\,\%$ is markedly higher than in the case without outliers.

[1] In Excel, $\hat{\mu}$ can be estimated with formula `=MITTELWERT()` (German) or `=MEAN()` (English).
[2] In Excel, $\hat{\sigma}$ can be estimated with formula `=STABW.N()` (German) or `=STDEV.P()` (English).

2.3.3 Step 3: Identifying Trend Breakers

Some of the observed percentage growth values can be considered outliers. These values often correspond to *trend breakers*, which are examined in the third step of the five-step process. Trend breakers represent disruptions to the smooth continuation of past developments and may arise from a variety of sources, such as economic crises, political changes, technological innovations, new discoveries, or shifts in market conditions.

Trend breakers occur more frequently than one might expect, as illustrated by the example of Zurich Airport. The collapse of Swissair in the early 2000s and the terrorist attacks of September 11, 2001, both had a strong impact on demand at Zurich. More recently, the COVID-19 pandemic and its profound influence on travel behavior caused another major disruption in the 2020s.

Although trend breakers play a decisive role in shaping system performance, they are notoriously difficult to identify in advance and even harder to predict. The literature therefore recommends *scenario analysis* as a method to explore potential disruptive events that could significantly affect a given system. In this course, however, we will not go into detail about modeling trend breakers. For our purposes, it is sufficient to recognize that they occur repeatedly and can have very substantial effects on engineering systems.

2.3.4 Step 4: Establishing Forecast (In)accuracy

The fourth step of the process is to evaluate the accuracy of historical forecasts for the factors subject to uncertainty. Since forecasts are often inaccurate, it is not surprising that large discrepancies frequently exist between predicted and actual outcomes. A useful way to visualize these discrepancies is with *porcupine graphs*, named for their resemblance to the spines of a porcupine. Figure 5 contains such a graph for St. Louis International Airport in the United States. It clearly illustrates how traffic forecasts made in 1998, 2001, 2002, 2003, and 2009 deviated substantially from actual traffic development at the airport.

Constructing porcupine graphs is often challenging, as many organizations do not systematically archive or make past forecasts publicly available. For Zurich Airport, however, two passenger demand forecasts could be retrieved: one from 2005 and another from 2015. Figure 6 plots the 2005 forecast as a black dashed line and the 2015 forecast as a red dashed line.

Excluding the COVID-19 outliers, these forecasts captured demand reasonably well for the period 2005–2019. Notably, they assumed average annual growth rates of 4.0% and 3.4%, respectively. These values are very close to the empirically estimated growth rate of 4.885% per annum, see Figure 4. We can therefore conclude that, at least for Zurich Airport, forecasts have been relatively reliable when compared to observed demand.

2.3.5 Step 5: Building a Dynamic Model

The fifth and final step of the process is to construct a **dynamic model** that mathematically captures the factors subject to uncertainty. Such a model provides not just a single forecast but a range of possible future outcomes, which we call *scenarios*. In the context of these scenarios, we focus on continuous-time random variables, i.e., variables that are defined at every point in time within a given interval. Typical examples of continuous-time random variables include demand, prices, sales figures, or revenues.

To model variables subject to uncertainty, we use a class of stochastic processes known as

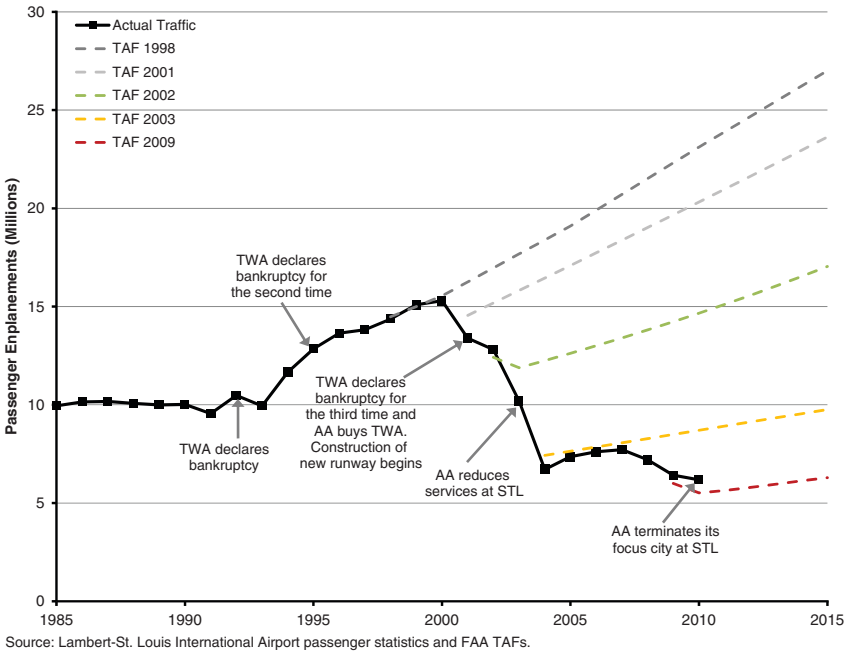

Figure 5: Example of porcupine graph for St. Louis International Airport. From Kincaid et al. (2012, p. 18).

diffusion models. In particular, we apply the **geometric Brownian motion** (GBM), a model widely used to describe uncertain growth processes (De Weck et al., 2007). While we omit the full mathematical derivation of GBM, we present here the core equations relevant for generating scenarios, in particular scenarios specifying the future levels of demand expected for an engineering system.

To create demand forecasts with GBM, we first have to define a planning horizon T, which specifies how far into the future the forecast extends. For example, in strategic planning a typical planning horizon is $T = 20$ years. Then, the time period between the start of the forecast and T is divided into discrete *planning phases* or *planning stages* t of equal duration, such that

$$t = 1, 2, \ldots, T, \tag{4}$$

where $t = 1$ denotes the first phase and $t = T$ the final phase at the end of the planning horizon.

Using GBM, the year-to-year change in demand ΔD_t can subsequently be expressed as

$$\Delta D_t = D_t \cdot \left(\hat{\mu} \cdot \Delta t + \hat{\sigma} \cdot \varepsilon \cdot \sqrt{\Delta t} \right), \tag{5}$$

where $\hat{\mu}$ is the estimated mean growth rate of demand, $\hat{\sigma}$ the estimated standard deviation of demand, $\varepsilon \sim \mathcal{N}(0, 1)$ a normally distributed random variable with zero mean and standard deviation of one, and Δt the time step (e.g., the duration of a single planning phase t). Note, Section 2.3.2 provides a detailed explanation on how to estimate $\hat{\mu}$ and $\hat{\sigma}$ on the basis of historical observations of demand.

Starting from an initial demand level D_0 which must be known, demand in the next period is obtained iteratively as

$$D_{t+1} = D_t + \Delta D_t. \tag{6}$$

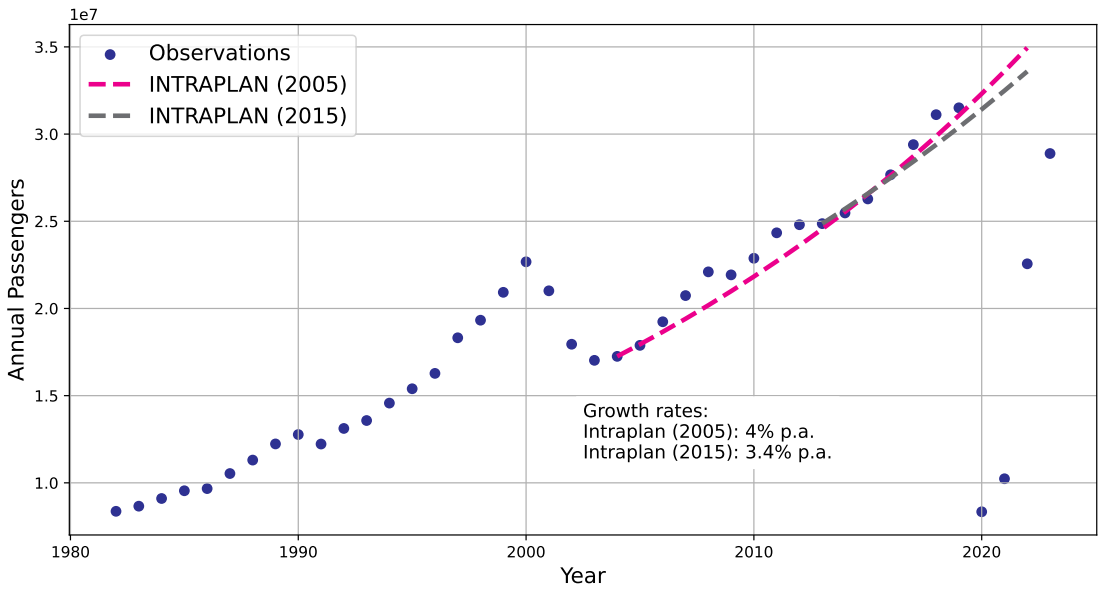

Figure 6: Porcupine graph for Zurich Airport

Applying this update rule expressed in Equation 6 successively across all planning phases t yields a complete **demand scenario** $\mathbf{D}^s$. In mathematical terms, a scenario is represented as a vector of demand levels across the planning horizon:

$$\mathbf{D}^s = [D_1^s, D_2^s, \ldots, D_T^s]\,, \tag{7}$$

where each element of the vector gives the forecasted demand in phase t for scenario s. To illustrate better, consider the example shown in Figure 7, which depicts one scenario for Zurich Airport generated with Equations 5–6, starting from an initial demand of $D_0 = 28.9$ million passengers per annum (MPPA). This scenario can be written as

$$\mathbf{D}^1 = [41.4, 47.4, 46.2, 59.6, 54.0, 55.5, \ldots, 111.7, 94.6] \text{ MPPA}. \tag{8}$$

This notation states, for example, that demand in the first planning phase is $D_1^1 = 41.4$ MPPA, while in phase $t = 3$ it is $D_3^1 = 46.2$ MPPA, and so on.

Because Equation 5 includes random variable ε, a different demand scenario $\mathbf{D}^s$ results each time the process described above is executed. Therefore, with GBM one can generate as many demand scenarios as required, resulting in a set of demand scenarios

$$\mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S \tag{9}$$

Figure 8 contains 30 randomly selected demand scenarios for Zurich Airport. Most of these scenarios remain close to the exponential growth trend (illustrated with a pink dashed line), while some diverge sharply upward or downward. This illustrates that the GBM-method can capture both strong growth (e.g., favorable economic conditions) and sharp declines (e.g., global recessions).

The key question is whether these scenarios are realistic? Evidence from the literature suggests that GBM approximates real-world (demand) growth processes reasonably well. To test this in practice, we conducted a backcasting experiment: assuming the year is 2004, we generated GBM-based demand scenarios for Zurich Airport starting at that point. As

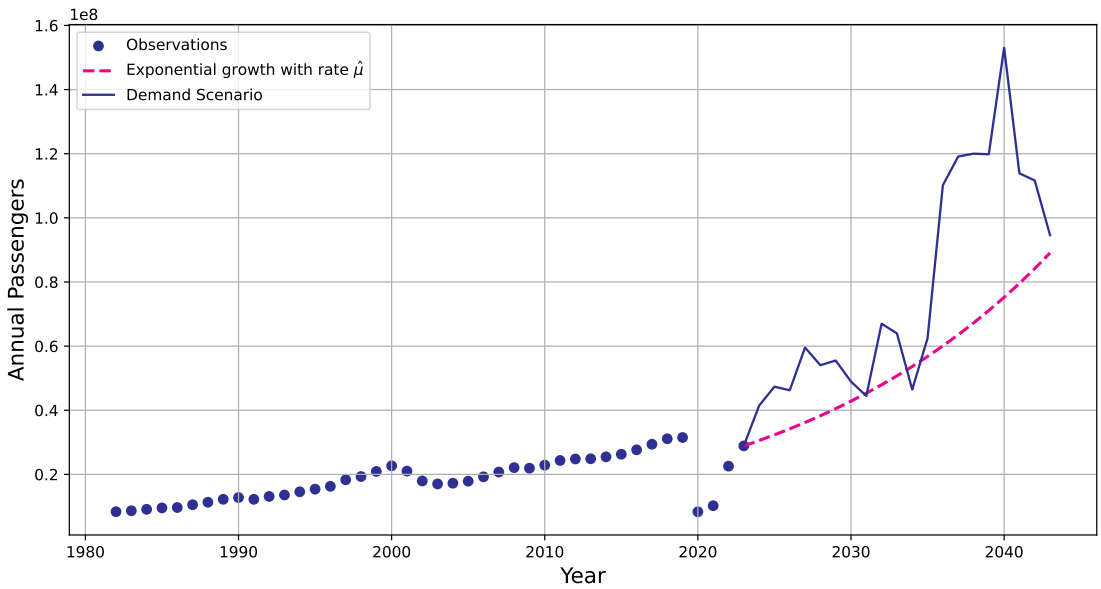

Figure 7: Example of GBM-based Demand Forecast for Zurich Airport

shown in Figure 9, the simulated scenarios broadly encompass the actual demand trajectory, including the sharp downturn during the COVID-19 pandemic (highlighted with pink dots). This result suggests that GBM-generated scenarios can realistically reflect both ordinary fluctuations and major disruptions (within certain limitations).

3 Real Options and Candidate Flexibilities

This section introduces the concept of real options and explains their relevance for engineering systems. We begin with financial options as a foundation, before distinguishing between real options *on* systems, which provide managerial flexibility, and real options *in* systems, which are embedded design features. Finally, we turn to the notion of *candidate flexibilities*, i.e., the potential ways in which such options can be identified and considered during system planning.

Learning Objectives

After working through Section 3,

- you can **describe** the term *real option*;
- you can **differentiate** between *real option on systems* and *real option in systems* and you can explain both terms; and
- for both real options on systems and real options in systems, you can **name** and **describe** three practical examples.

3.1 Options in Finance

In order to understand the idea of real options, it is helpful to first review the concept of a financial option. Unlike in everyday English, where an *"option"* simply means an alternative or possibility, in finance an option is a specific contract that grants the right—but not the obligation—to take a predetermined action in the future at a predetermined cost. More specifically, a financial option is an agreement between a buyer and a seller that specifies

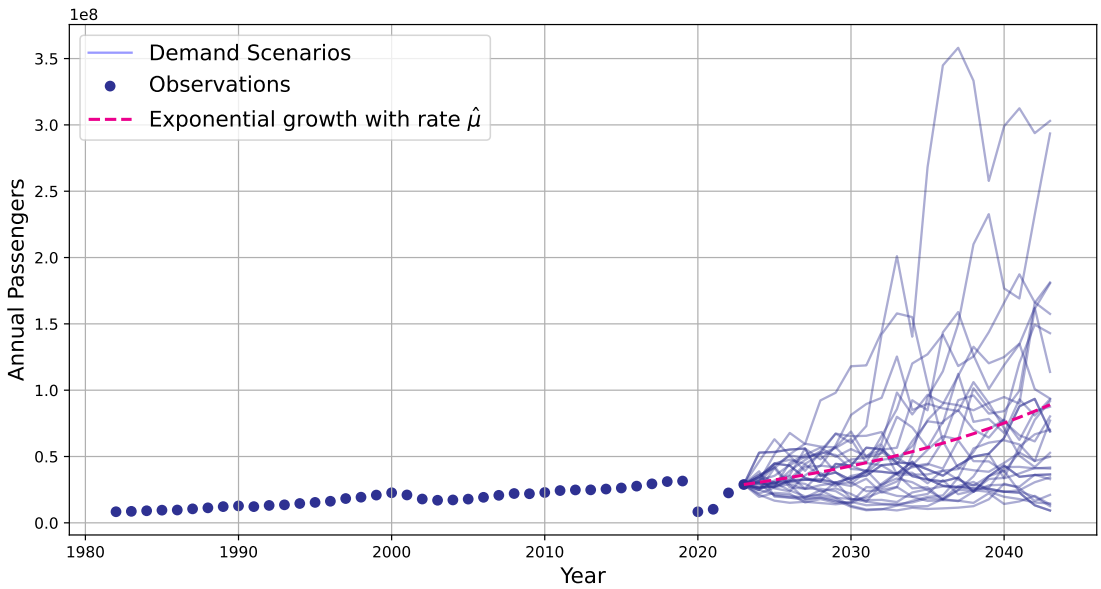

Figure 8: GBM-based Demand Forecasts for Zurich Airport (30 Randomly Selected Scenarios)

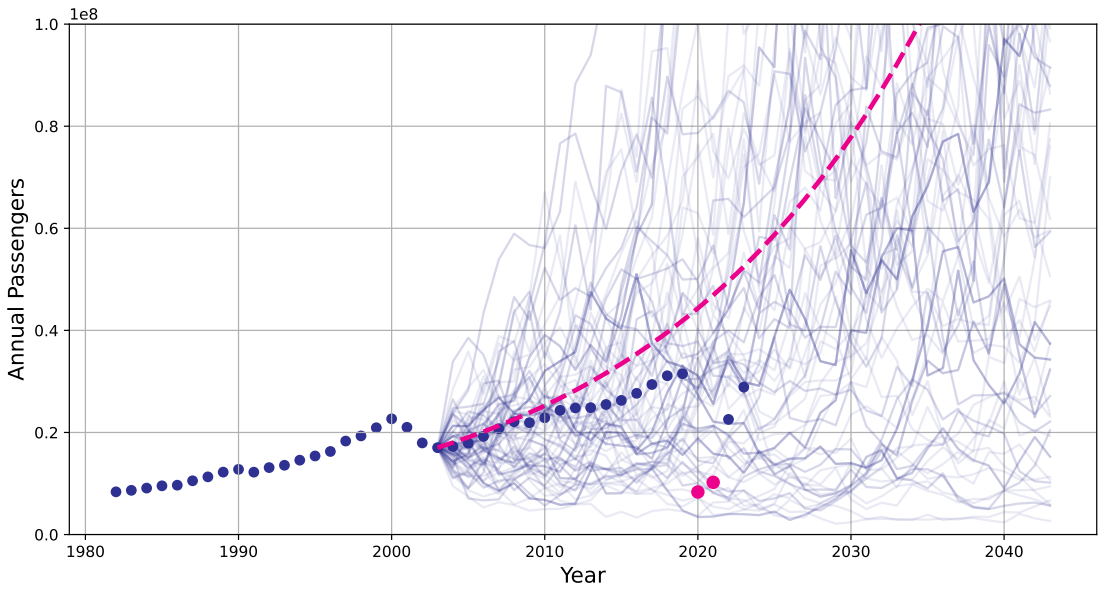

Figure 9: Thirty GBM-based demand scenarios for Zurich Airport (starting in 2004).

in advance the terms under which the buyer may exercise the option. To obtain this right, the buyer pays the seller a fee, called the premium. In finance, two basic types of options exist: the call option and the put option.

- A **call option** gives the holder the right, but not the obligation, to buy an underlying asset (such as a share or commodity) at a predetermined exercise or strike price at some future point in time—even if the market price has risen above that strike price.
- A **put option**, in contrast, gives the holder the right, but not the obligation, to sell the asset at the strike price—even if the market price has fallen below that level.

To see the advantage that options can provide, let us examine a **call option** under two

possible future scenarios:

1. *Market price less than or equal to the strike price*: At maturity (i.e., the point in time the option can be exercised), if the market price of the asset is less than (or equal to) the strike price, exercising the call option would mean buying the asset for more than it is currently worth. Since this would generate a loss, the rational choice is simply to let the option expire worthless. The only cost to the holder is the premium that was paid upfront.

2. *Market price greater than the strike price*: If, instead, the market price at maturity is higher than the strike price, the option holder can purchase the asset at the lower strike price and immediately realize a gain. The payoff is the difference between the market price and the strike price. Once this payoff exceeds the premium originally paid, the option generates a net profit, see Figure 10.

Similarly, the following two scenarios are to be considered for a **put option**:

1. *Market price greater than or equal to the strike price*: At maturity, if the market price of the asset is higher than (or equal to) the strike price, exercising the put option would mean selling the asset for less than it is currently worth. Since this would generate a loss, the rational choice is simply to let the option expire worthless. The only cost to the holder is the premium that was paid upfront.

2. *Market price less than the strike price*: If, instead, the market price at maturity is lower than the strike price, the option holder can sell the asset at the higher strike price. The payoff is the difference between the strike price and the market price. Once this payoff exceeds the premium originally paid, the option generates a net profit, see Figure 10.

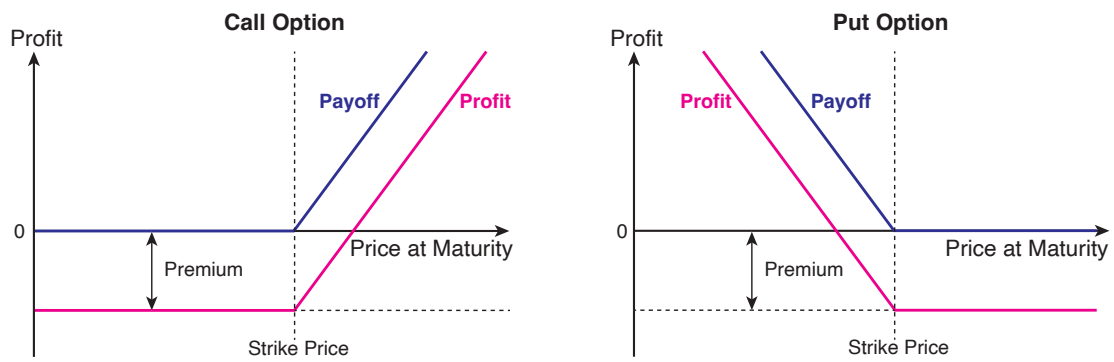

Figure 10: Profit vs. Price At Maturity for Call Option (left) and Put Option (right)

In summary, call and put options serve distinct purposes. Call options are primarily used to capitalize on potential opportunities: they allow the holder to benefit from favorable upward movements in the price of an asset while keeping the maximum possible loss limited to the premium. Put options, on the other hand, are often employed to avert risks of negative developments: they act as a form of insurance by gaining value when the price of the underlying asset declines, thereby offsetting potential losses.

3.2 Real Options

After this brief detour into the world of finance, we now turn to the concept of the real option. Unlike financial options, which are contracts on traded assets, real options concern

engineering systems (De Neufville, 2003). A real option gives its owner the right, but not the obligation, to modify a system in response to uncertainty. In other words, real options embed managerial flexibility directly into the planning and operation of engineering systems (Brealey et al., 2022; Cardin, Zhang, & Nuttall, 2017). Similar to their financial counterparts, real options mitigate downside risk when conditions turn unfavorable while preserving the ability to capitalize on upside opportunities. Empirical studies suggest that engineering systems designed with real options are up to 30% more valuable than conventionally planned engineering systems (i.e., systems not containing real options) (Cardin, 2014; Geltner & De Neufville, 2018; Schwartz & Trigeorgis, 2004).

The literature distinguishes between two categories of real options. On the one hand, **real options on systems** represent sources of managerial flexibility, allowing owners to buy, sell, expand, or contract an engineering system without being bound to a fixed course of action. On the other hand, **real options in systems** are physical design features intentionally built into the system to enable future adaptations (Wang & De Neufville, 2005). Put differently, real options on systems operate as management levers external to the system, while real options in systems are embedded capabilities that alter the system's design and performance over time.

3.2.1 Real Options on Systems

With the definitions of real options *on* and *in* systems in mind, we now turn to the generic real option types that literature has identified. Trigeorgis (Trigeorgis, 1996) and Trigeorgis and Reuer (Trigeorgis & Reuer, 2017) discuss six fundamental categories of real options. It is important to note that while these types are conceptually general, their applicability varies across different domains. In particular, not all of them find direct use in the strategic planning of large-scale engineering systems.

- **Option to defer.** This option grants decision makers the right to postpone an investment until uncertainty is resolved or conditions become more favorable. For engineering systems such as airport terminals, railway stations, or power plants, deferral options are highly relevant since demand forecasts or regulatory developments are often uncertain. Waiting can thus protect against premature investment in oversized or misaligned infrastructure.
- **Option to stage.** Also referred to as a sequencing or phased-investment option, this allows a project to be structured as a series of smaller, incremental commitments rather than a single, irreversible one. Staging reduces downside exposure while keeping the upside open. In engineering systems, staging is frequently applied: railway stations or airport terminals are often expanded in phases, with each stage contingent on observed traffic growth.
- **Option to grow.** Growth options arise when an initial investment creates opportunities for future expansion. For example, investing in a basic facility might pave the way for adding new services, technologies, or complementary infrastructure. While growth options are more prominent in R&D-intensive sectors, they also apply to engineering systems.
- **Option to alter scale.** This option provides the flexibility to expand, contract, or temporarily suspend operations depending on conditions. In manufacturing, this often takes the form of outsourcing or flexible production capacity. For engineering systems, the application is more limited, since scale adjustments (e.g., resizing a bridge or terminal) are technically and economically challenging. Still, operational

scaling (e.g., opening or closing parts of a railway station) can reflect this option in practice.

- **Option to switch.** Switching options allow decision makers to change the mode of operation, inputs, or outputs depending on relative prices and availability. In process industries, this could mean switching fuels or suppliers. For engineering systems, examples include power plants with multi-fuel capability, or airports switching terminal usage between domestic and international traffic.

- **Option to abandon.** The abandon option gives the right to exit a project and recover some salvage value. While highly relevant in financial and industrial investments, its application to engineering systems is limited. Infrastructure such as bridges, railway stations, or airports rarely has salvage value. However, abandonment may take the form of repurposing (e.g., converting an old industrial site into a transport hub) or partial decommissioning with limited salvage value.

In summary, among these six types, the *option to defer* and the *option to stage* are particularly important for engineering systems, functioning much like call options that preserve the right to invest later or incrementally. The *option to abandon* resembles a put option, though its usefulness is often constrained by low salvage values of infrastructure assets. Finally, it is worth noting that in practice, real options are seldom exercised in isolation. Engineering system projects frequently embed multiple option types simultaneously, for instance combining staging with growth opportunities or integrating deferral and abandonment flexibility into one *candidate flexibility*, see Section 3.3.

3.2.2 Real Options in Systems

Real options *in* systems are design features deliberately built into engineering systems to enable future changes in size or functionality (Wang & De Neufville, 2005). Unlike real options *on* systems, which can be applied without detailed technical insight, the design, planning, and implementation of real options *in* systems require in-depth engineering knowledge. Because these features must be tailored to each system, generic templates are rare; instead, the literature documents a variety of illustrative real-world cases. The following examples highlight how flexibility can be embedded in practice:

- One well-known case is the parking garage of the Blue Water Shopping Center in the UK (De Neufville et al., 2006). At the time of construction, future parking demand was highly uncertain. To accommodate this, the garage was initially built with four levels, but the structure and foundations were deliberately oversized to support up to four additional floors. Although this upfront investment made the initial construction relatively costly, it enabled rather inexpensive expansion later, once demand justified it, thereby improving the project's long-term value.

- A similar principle guided the design of the *Bentall 5 office tower* in Vancouver, Canada (Geltner & De Neufville, 2018; Guma et al., 2009). Initial demand for office space did not warrant a full-height tower, so only half the potential floors were constructed. However, the building was structurally prepared for vertical expansion, which was successfully realized several years after initial completion.

- Real options in systems are not limited to expansion in size; they can also adapt system functionality. A notable example is the *25 April Bridge* in Lisbon, Portugal (Chambers, 2007). Originally built in the 1960s as a motorway bridge, it was structurally designed to allow for the later addition of a second deck. This option was

eventually exercised, enabling rail traffic to cross the bridge and thereby adding a new function to the existing infrastructure.

These cases demonstrate how embedding flexibility at the design stage can create valuable adaptation opportunities for uncertain futures. While costly upfront, such provisions often pay off by reducing the risks of under- or over-design and by enabling systems to evolve alongside demand and other factors.

3.3 Candidate Flexibilities

In the context of real options for engineering systems, the literature frequently refers to so-called *candidate flexibilities* (Cardin, Zhang, & Nuttall, 2017; Wang & De Neufville, 2005). A candidate flexibility is a potential way of introducing flexibility into an engineering system through real options. In practice, there are typically several alternatives for doing so: flexibility can be embedded *in* the system's design (e.g., modular or expandable components) or exercised *on* the system through managerial choices (e.g., deferring, staging, or abandoning investments).

Consider a fictitious office building. There are multiple ways in which this system could be made flexible:

- *Candidate Flexibility A:* Apply the *option to defer*. For instance, reserve adjacent land through *land banking*, thereby creating the possibility of expanding the building at a later, more favorable time.
- *Candidate Flexibility B:* Apply the *option to stage*. This could be realized by planning the building in successive horizontal expansion phases, with each stage contingent on future demand.
- *Candidate Flexibility C:* Introduce a real option in system allowing the vertical expansion of the building at a later point in time (similar to the *Bentall 5 office tower* in Vancouver, which incorporated structural features to enable additional floors to be added after initial construction)

Identifying candidate flexibilities is an early step in the design and planning process. At this stage, the goal is to recognize and enumerate possible options without committing to their implementation. Once identified, these candidate flexibilities must be evaluated in terms of feasibility, cost, and expected value. The subsequent task is to determine which candidate flexibilities are worth realizing. Methods for evaluating and selecting among candidate flexibilities are discussed in Sections 4 and 6.

4 Evaluation of Conventional Facility Requirements

In this section, we introduce how conventionally planned engineering systems can be evaluated. We first define the concept of capacity and explain how conventional facility requirements are expressed through capacity vectors. We then show how the economic value of conventional facility requirements can be assessed using the Net Present Value (NPV) and its extension, the Expected Net Present Value (ENPV).

Learning Objectives

After working through Section 4,

- you can **describe** what *conventional facility requirements* are and how they can be represented with a capacity vector;
- on the basis of future costs and revenues, you can **calculate** the net present value (NPV) and the expected net present value (ENPV) of a conventionally planned engineering system;
- you can **explain** why costs or revenues incurred at different times need to be discounted for the NPV calculation (*time value of money*);
- you can **model** revenues and costs that arise in engineering systems using simple mathematical functions; and
- you can explain the term *economies of scale effects* and you can **calculate** economies of scale effects using a power cost function.

4.1 What is Capacity?

Capacity refers to the maximum level of output or service that a system, facility, or piece of equipment can deliver under normal operating conditions within a specified period of time. In other words, it represents the upper bound on the amount of demand, whether production, flow, or service, that can be met.

For example, in a bakery, capacity refers to the maximum number of loaves of bread that can be produced per day. At an airport, it may denote the maximum number of flights or passengers that can be processed within a given unit of time. In the case of a power plant, capacity corresponds to the maximum amount of power, measured in megawatts (MW), that can be generated or transmitted, etc.

4.2 Conventional Facility Requirements

Facility requirements specify how, when, and, if relevant, where the capacity of an engineering system must be adjusted over time in order to meet future demand levels. For engineering systems that are planned in the conventional sense, meaning without considering real options, conventional facility requirements are to be defined. They can be understood as *schedules* or *timetables* that define the timing and manner in which the system's capacity is adapted or expanded in the future.

As already discussed in Section 2, a strategic plan is always drawn up for a certain **planning horizon** T, which specifies the length of time into the future for which a plan or strategy is developed, extending from the present moment to a defined future point. The planning horizon specifies how far ahead decisions are made and resources are allocated. Therefore, conventional facility requirements are expressed through a capacity vector $\mathbf{K}$ containing T elements. As introduced in Equation 10, this vector represents the *capacity deployment schedule* of the engineering system over the entire planning horizon:

$$\mathbf{K} = [K_1, K_2, \ldots, K_T]\,, \tag{10}$$

where each element K_t of the vector specifies the available capacity in planning phase t. For example, K_1 denotes the capacity in the first phase $t = 1$, K_2 in the second phase $t = 2$, and so forth, up to K_T, which represents the capacity available in the final planning phase.

Figure 11 illustrates a practical example of conventional facility requirements for the runway system of a fictitious airport. The x-axis represents time, while the y-axis indicates the runway capacity, measured in number of air traffic movements (ATM) per hour it can handle. The figure shows three alternative facility requirements, denoted by $\mathbf{K}_1$ (blue), $\mathbf{K}_2$ (pink), and $\mathbf{K}_3$ (grey). Irrespective of the chosen facility requirement alternative, the initial capacity of the runway system is $K_0 = 28$ movements per hour.

For instance, facility requirement $\mathbf{K}_1$ can be written as

$$\mathbf{K}_1 = [28, 28, 34, 34, 34, 38, 38, 38, 38, 38] \,, \tag{11}$$

where each entry specifies the capacity in the corresponding planning stage. In this example, the system's capacity is $K_1 = 28$ movements per hour at planning phase $t = 1$, increases to 34 movements at $t = 3$, and reaches 38 movements at $t = 6$, where it remains constant until $t = 10$.

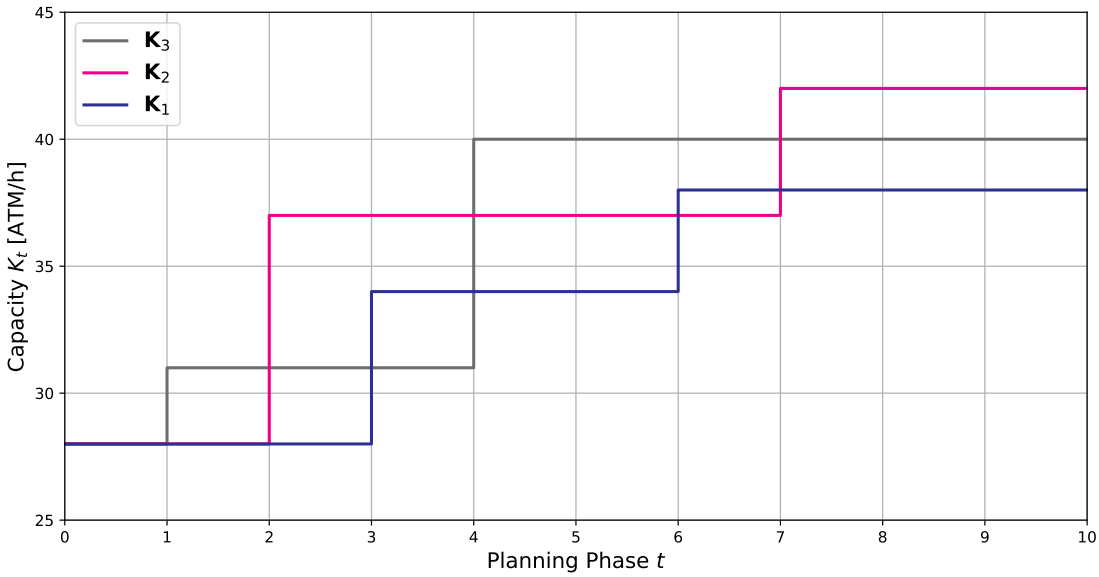

Figure 11: Conventional Facility Requirements for the Runway System of a Fictitious Airport

4.3 Evaluation of Conventional Facility Requirements

Having defined conventional facility requirements by means of a capacity vector $\mathbf{K}$, the next step is to consider how the *quality* or *suitability* of a facility requirement can be evaluated in an objective manner. While the literature mentions various ways to define and measure the quality of a facility requirement, we restrict our attention to economic evaluation methods, with the goal of determining the economic value of an engineering system planned with conventional facility requirements $\mathbf{K}$.

The literature discusses several approaches and measures for economic evaluation of facility requirements for engineering systems, including the *net present value*, the *rate of return*, and *cost–benefit ratios*, among others. For a comprehensive overview of these methods, we refer to Remer and Nieto (1995a, 1995b). In the following, however, we focus exclusively on the application of the *Net Present Value* (NPV).

Inputs Required for Evaluation. Having selected an evaluation technique, we can now turn our attention to the general process of assessing facility requirements for a given engineering system. This process requires two inputs, namely (i) an inventory and (ii) one or more forecasts specifying the expected future demand levels of the system. The **inventory** includes information such as the capacity K_0 of the engineering system in available at $t = 0$ as well as data describing the system's overall performance. These performance measures are highly specific and depend on the engineering system in question. For instance, a typical performance measure of a bakery is the hourly "production rate" of one single oven, for a power plant it is the maximum continuous output power of a generator, etc.

The second input required to assess facility requirements is information on the expected levels of demand for the future. As discussed in Sections 1 and 2, one single **demand forecast** scenario $\mathbf{D}^s$ is typically created in the context of conventional strategic planning, while multiple (i.e. thousands) scenarios $\mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S$ are created for flexible strategic planning purposes.

Discounted Cash Flow Method. Having the inventory of the system as well as demand scenarios available, one can conduction the evaluation of facility requirements $\mathbf{K}$ in a second step. To this end, one must to calculate the **Net Present Value** (NPV) of the system given facility requirement $\mathbf{K}$, demand forecast $\mathbf{D}^1$ (or demand forecasts $\mathbf{D}^1, \mathbf{D}^2, \ldots$), and other factors specified in the inventory, if required. The NPV is usually determined with the *discounted cash flow method*, which is defined as

$$NPV = -\mathcal{C}_0 + \sum_{t=1}^{T} \frac{1}{(1+\delta)^t} \cdot (\mathcal{R}_t - \mathcal{C}_t) \tag{12}$$

where $\mathcal{R}_t$ are revenues generated by the system in planning phase t, $\mathcal{C}_t$ are costs incurred by the system in planning phase t, $\mathcal{C}_0$ are initial costs incurred at $t = 0$, and $\delta \geq 0$ is the discount rate.

As the name *discounted cash flow method* suggests, the NPV is defined as the sum of the expected future cash flows of the engineering system, discounted to their present value. A **cash flow** refers to the movement of money into and out of a business, a project, or, in our case, an engineering system over a given period of time, e.g. within a planning phase t. In engineering systems, inflows typically arise from revenues generated by the system, while outflows represent expenses, operating costs, investments, or debt repayments. For our purposes, we assume that revenues contribute positively to the NPV (and are therefore added in Equation 12), whereas costs contribute negatively (and are therefore subtracted in Equation 12).

Time Value of Money The cash flows in Equation 12 occur at different points in time, namely the planning stages $t = 1, 2, \ldots, T$. Monetary values at different times cannot be directly compared with each other because the value of money changes over time. As De Neufville (1990, p. 205) explains: "A dollar now is worth more than a dollar in the future because it can, in general, be used productively between now and then. At a personal level, for example, you can place money in a savings account and get a larger amount back after a while. In the economy at large, businesses and governments can use money to build things, grow food, educate people, and do other worthwhile activities." This principle, is known as the *time value of money*.

To make cash flows V_t that occur at different points in time t comparable with each other, they must be **discounted** to their **present value** PV using the following formula:

$$PV = V_t \cdot \frac{1}{(1+\delta)^t} \quad \text{with } \delta \geq 0, t = 1, 2, \ldots, \tag{13}$$

where δ is the discount rate, and t refers to the points in time (e.g., number of years from now).

The **discount rate** δ reflects the time value of money, capturing both the opportunity cost of capital (i.e., the return that could be earned on an alternative investment) and, in some cases, an adjustment for inflation or risk. As shown in Figure 12, the choice of the discount rate δ strongly influences the present value PV of a cash flow over time. The figure plots the present value PV of a cash flow of $V_t = 100$ on the y-axis against time t on the x-axis, using discount rates of $\delta \in \{0.02, 0.05, 0.10\}$. A lower discount rate results in a slower decline of the present value over time, meaning that future cash flows retain a relatively high present value compared to immediate ones. Conversely, a higher discount rate accelerates the decline, so that future cash flows are worth substantially less in present value terms. For example, the present value of $V_t = 100$ at time $t = 0$ is, by definition, CHF 100. However, if CHF 100 is to be received in ten years at a discount rate of $\delta = 0.10$, its present value decreases to approximately CHF 39. This means that to cover a payment of CHF 100 in ten years, it is sufficient to invest only CHF 39 today if the available capital yields an annual return of 10%; the remaining CHF 61 will be accumulated through interest earned over the intervening years.

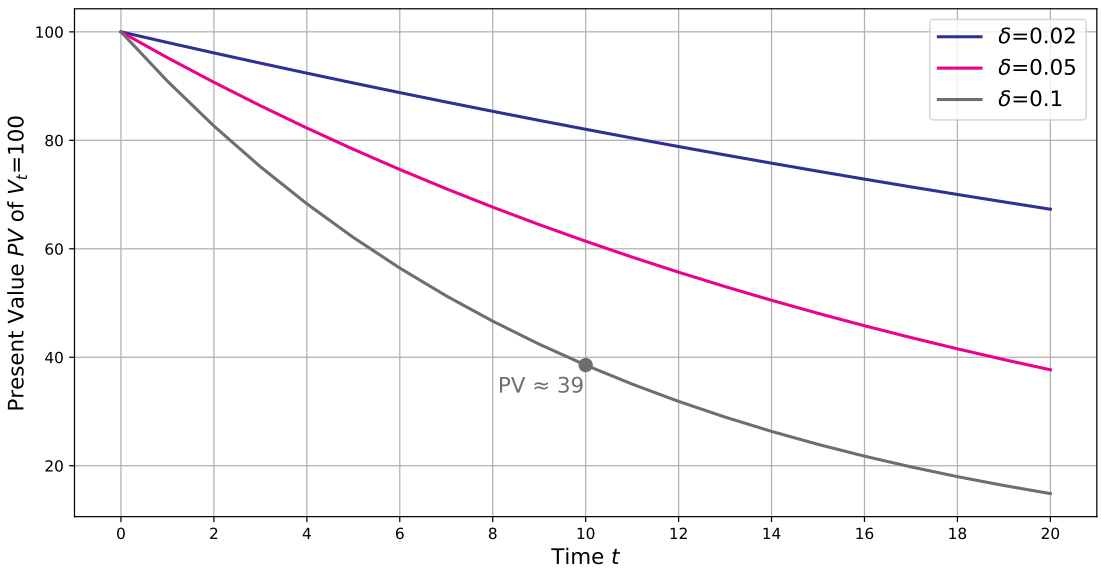

Figure 12: Present Value vs. Time

This raises the question of which discount rate should be applied to a project or an engineering system? Unfortunately, there is no single definitive answer, as the choice of an appropriate discount rate depends on a variety of factors and circumstances. As a general rule, the higher the risk of an investment (i.e., the greater the probability that it will not yield the expected returns), the higher the discount rate should be. In addition, the choice of the discount rate strongly depends on the capital structure of the project's financiers. Companies and governments therefore often refer to the so-called weighted average cost of capital (WACC) when determining the discount rate. For infrastructure projects, a reasonable discount rate is typically in the single-digit percentage range. However, to obtain

Planning phase t	Project 1		Project 2		Project 3	
	$\mathcal{R}_t$	$\mathcal{C}_t$	$\mathcal{R}_t$	$\mathcal{C}_t$	$\mathcal{R}_t$	$\mathcal{C}_t$
0	0	1000	0	1000	0	1000
1	700	200	500	300	500	210
2	600	200	600	300	600	210
3	500	200	700	300	700	210

Table 1: Expected Revenues and Costs of Projects 1, 2, and 3

a conclusive answer for the rate applicable in a specific case, it is advisable to consult the finance department of the respective company or government.

Net Present Value. Now that we understand how to discount cash flows, we return to the definition of the Net Present Value (NPV) in Equation 12. To calculate the NPV of an engineering system under conventional facility requirements $\mathbf{K}$, all expected future revenues and costs are to be discounted to their present value and summed. Finally, the initial costs $\mathcal{C}_0$, which may arise, for instance, from initial construction activities, are subtracted from this sum.

Assume we have three different facility requirements $\mathbf{K}_1$, $\mathbf{K}_2$, and $\mathbf{K}_3$ for a fictitious engineering system, denoted as *Project 1*, *Project 2*, and *Project 3*. Moreover, we consider a planning horizon of $T = 3$, a discount rate of $\delta = 0.05$, as well as revenues $\mathcal{R}_t$ and costs $\mathcal{C}_t$ as specified in Table 1. In this case, the resulting NPV of the system under facility requirement $\mathbf{K}_1$, i.e., *Project 1*, is equal to

$$
\begin{aligned}
NPV_1 &= -\mathcal{C}_0 + + \frac{1}{(1+\delta)^1}(\mathcal{R}_{1,1} - \mathcal{C}_{1,1}) + \frac{1}{(1+\delta)^2}(\mathcal{R}_{2,1} - \mathcal{C}_{2,1}) + \frac{1}{(1+\delta)^3}(\mathcal{R}_{3,1} - \mathcal{C}_{3,1}) \\
NPV_1 &= -1000 + \frac{1}{(1.05)^1}(700 - 200) + \frac{1}{(1.05)^2}(600 - 200) + \frac{1}{(1.05)^3}(500 - 200) \\
NPV_1 &= \text{CHF } 98.2, \qquad (14)
\end{aligned}
$$

while for projects 2 and 3, it is $NPV_2 = \text{CHF } -191.9$ and $NPV_3 = \text{CHF } 53.2$, respectively.

Interpretation of NPV-Results. After inspecting the resulting NPVs of Projects 1, 2, and 3, two questions arise: (i) what are the implications of positive or negative NPVs, and (ii) which of the projects is the most favorable?

As a general rule, **projects with a positive NPV are economically viable**, while projects with a negative NPV destroy value and should not be pursued. A positive NPV means that the discounted value of future inflows exceeds the discounted value of outflows, thereby generating an economic surplus and increasing the wealth of the investor. Conversely, a negative NPV indicates that inflows are insufficient to cover the costs in present-value terms, leading to a destruction of value. In such cases, the investor is better off allocating resources elsewhere or not investing at all.

When comparing multiple alternatives, the best project is the one resulting in the highest NPV, as it provides the greatest net benefit. Accordingly, when selecting the most suitable facility requirement for an engineering system, the option that **maximizes the system's**

NPV should be chosen. Comming back to the three projects mentioned before: one should select Project 1, as it results in the highest NPV of the engineering system among all projects evaluated. It should be noted, however, that particularly in the infrastructure sector it is quite common for projects to have negative NPVs, as many systems generate little or no direct revenues at all. Nevertheless, even in these cases, the facility requirement which maximizes the NPV of the system should be selected.

Modeling Revenues and Costs. As illustrated in the previous example, the calculation of the NPV of an engineering system under facility requirements $\mathbf{K}$ requires knowledge of both of the expected future revenues $\mathcal{R}_t$ generated and costs $\mathcal{C}_t$ incurred by the system at each planning phase $t = 1, \ldots, T$. In Table 1, we assumed that these values are known. In practice, however, this assumption does not hold. Instead, methods are required to estimate the future expected revenues and costs of an engineering system.

Both revenues and costs depend on the future capacity $\mathbf{K}$ of the system (i.e., the facility requirements) as well as on the expected future demand levels $\mathbf{D}^s$. Accordingly, Equation 12 can be extended as follows:

$$NPV^s(\mathbf{K}, \mathbf{D}^s) = -\mathcal{C}_0 + \sum_{t=1}^{T} \frac{1}{(1+\delta)^t} \cdot \big(\underbrace{\mathcal{R}_t(K_t, D_t^s)}_{\text{Revenues}} - \underbrace{\mathcal{C}_t(K_t, \Delta K_t, D_t^s)}_{\text{Costs}} \big), \tag{15}$$

where $\mathcal{R}_t(K_t, D_t^s)$ and $\mathcal{C}_t(K_t, \Delta K_t, D_t^s)$ denote the revenues generated and costs incurred in planning phase t under facility requirements $\mathbf{K}$ and demand scenario $\mathbf{D}^s$. Note that costs also depend on ΔK_t, which describes whether, and by how much, the capacity of the engineering system is adjusted in planning phase t: If $\Delta K_t = 0$, the capacity remains unchanged; if $\Delta K_t > 0$, the capacity is increased; and if $\Delta K_t < 0$, the capacity is reduced.

The **revenues** $\mathcal{R}_t$ generated by an engineering system in planning phase t depend usually on both the demand for the system D_t^s and its capacity K_t. Revenues can take many forms and are unique to each system. For example, if a system generates both rental income $\mathcal{R}_{\text{Rental},t}$ and sales revenues $\mathcal{R}_{\text{Sales},t}$, the total revenues $\mathcal{R}_t$ can be expressed with the following sum

$$\mathcal{R}_t = \mathcal{R}_{\text{Rental},t} + \mathcal{R}_{\text{Sales},t}. \tag{16}$$

For every identified source of revenue, a mathematical relationship must be defined that specifies the expected magnitude of these revenues. To illustrate better, consider the example of a large office building whose capacity is measured in terms of the available office space. Suppose that at time t, the entire capacity is rented out to customers. In this case, the resulting rental revenues $\mathcal{R}_{\text{Rental},t}$ can be expressed with

$$\mathcal{R}_{\text{Rental},t} = K_t \cdot r_{\text{Rental}}, \tag{17}$$

where r_{Rental} denotes the rental income generated per square meter of office space.

In practice, demand is often not equal to available capacity (it is either lower or higher than capacity). In such cases, Equation 17 must be extended as follows:

$$\mathcal{R}_{\text{Rental},t} = \begin{cases} D_t^s \cdot r_{\text{Rental}}, & \text{if } D_t^s \leq K_t, \\ K_t \cdot r_{\text{Rental}}, & \text{if } D_t^s > K_t. \end{cases} \tag{18}$$

Equation 18 is to be understand as follows: If demand is lower than system capacity (i.e., $D_t^s \leq K_t$), rental revenues correspond to the product of demand D_t^s and r_{Rental}. However, if demand exceeds capacity (i.e., $D_t^s > K_t$.), only as much space can be rented as is available, and rental revenues are limited to the product of capacity K_t and r_{Rental}.

Costs $\mathcal{C}_t$ incurred by an engineering system in planning phase t usually arise either from operation (e.g., cleaning, staffing), or investments in capacity (e.g., the construction of additional facilities). Similar to revenues discussed above, costs depend on the demand for the system D_t^s and its capacity K_t. Additionally, to determine investment costs, the change in capacity ΔK_t must be considered as well.

The mathematical formulation of operating and maintenance costs normally follows the same structure as outlined in Equations 17 and 18. In the case of operating costs, a distinction is often made between capacity-driven and demand-driven cost-components. Returning to the office building example, heating costs are largely capacity-driven, since all building areas must be heated regardless of occupancy. In contrast, cleaning costs are strongly demand-driven, as they depend mainly on the actual use of the facility. For this example, Equation 18 can be adapted as follows to estimate the operating costs of the office building:

$$\mathcal{C}_{\text{Operation},t} = \begin{cases} K_t \cdot c_{\text{Heating}} + D_t^s \cdot c_{\text{Cleaning}}, & \text{if } D_t^s \leq K_t, \\ K_t \cdot c_{\text{Heating}} + K_t \cdot c_{\text{Cleaning}}, & \text{if } D_t^s > K_t, \end{cases} \tag{19}$$

where c_{Heating} denotes the heating costs per square meter of office space, and c_{Cleaning} the cleaning costs per square meter of office space.

When determining investment costs (also referred to as *installation costs*) the literature advises using a different modeling approach, because these costs typically benefit from **economies of scale** effects (Luss, 1982; Martínez-Costa et al., 2014; Van Mieghem, 2003). Economies of scale describe cost advantages that arise because the average cost per unit of capacity $c_{\text{Installation}}$ typically decreases with the expansion size ΔK_t. A simple analogy can be found in online shopping: buying a single pair of shoes comes at a fixed price, whereas purchasing several pairs often results in a discount, thereby reducing the average price per pair. The same principle applies to engineering systems: small capacity adjustments yield only limited savings, while larger adjustments can realize significant cost advantages.

To model economies of scale, a *power cost function* can be used. For this purpose, let $c_{\text{Installation}}$ denote the cost per one single unit of capacity, e.g., the cost of one oven. The total installation costs of an adjustment of ΔK_t units of capacity can then be expressed as

$$\mathcal{C}_{\text{Installation}} = (\Delta K_t)^{\alpha} \cdot c_{\text{Installation}}, \tag{20}$$

where $0 \leq \alpha \leq 1$ is the *economies of scale factor*, which specifies the strength of the effect. For $\alpha = 1$, no economies of scale are realized. The closer α is to zero, the stronger the economies of scale become. In practical applications, α typically lies between 0.6 and 1 (Cardin & Hu, 2016).

Figure 13 illustrates how different economies of scale values α affect the installation costs for a capacity adjustment ΔK, assuming $c_{\text{Installation}} = 1$. The grey line depicts costs without economies of scale effects, where the total cost increases linearly with ΔK (e.g., 20 units cost 20, 40 units cost 40, and so on). By contrast, under economies of scale (blue and pink lines), the average cost per unit decreases as the size of the capacity adjustment increases.

For example, the average cost per unit is higher when adjusting by 20 units compared to adjusting by 40 units.

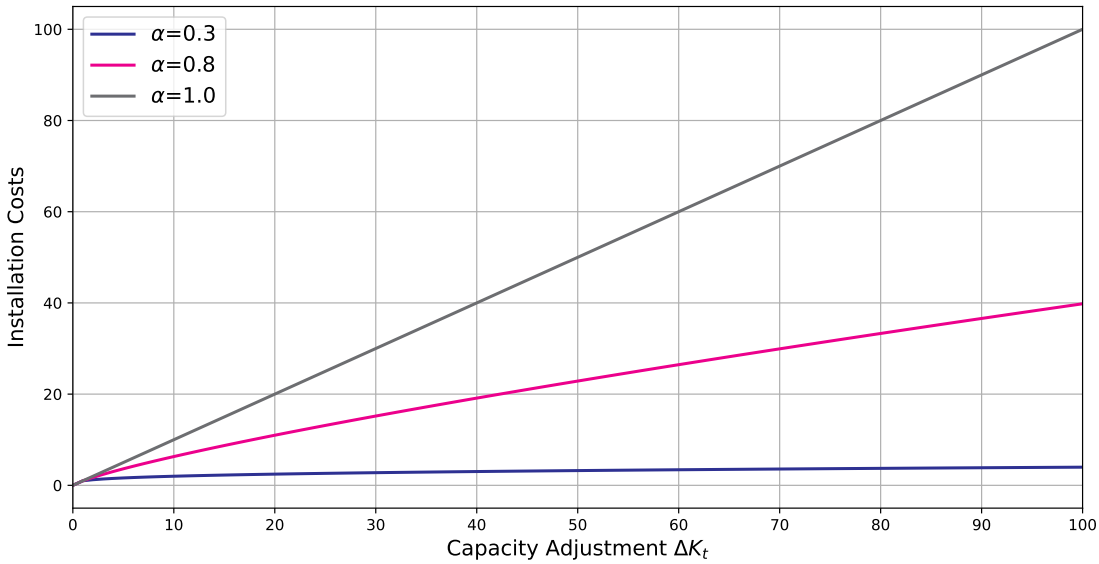

Figure 13: Illustration of economies of scale for different values of α.

Expected Net Present Value Equation 12 defines how the NPV of a system can be determined under a conventional facility requirement $\mathbf{K}$ and one specific demand scenario $\mathbf{D}^s$. However, as discussed in Section 2, planners are advised to evaluate strategic plans for engineering systems not only with respect to one single demand scenario $\mathbf{D}^s$, but rather by considering a large number of scenarios $\mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S$. As such, to properly account for uncertainty, the literature recommends considering between 2000 and 10000 demand scenarios (Geltner & De Neufville, 2018).

When more than one demand scenario is included in the evaluation of an engineering system, the Expected Net Present Value (ENPV), which is the expected value of the NPV across all scenarios, must be determined. Assuming that the probability of occurrence p_s of all scenarios is equal (i.e., $p_1 = p_2 = \ldots, p_S$) and $\sum_{s=1}^{S} p_s = 1$, the ENPV of an engineering system under conventional facility requirement $\mathbf{K}$ and demand scenarios $\mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S$ is defined as

$$ENPV(\mathbf{K}, \mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S) = \sum_{s=1}^{S} p_s \cdot \underbrace{NPV(\mathbf{K}, \mathbf{D}^s)}_{\text{see Equation 12}}. \tag{21}$$

5 Evaluation of Flexible Facility Requirements

This section introduces the concept of *flexible facility requirements*. We first motivate why conventional facility requirements (i.e., capacity vectors) are insufficient when real options are available, and then show how decision rules can be used to capture flexibility. Finally, we illustrate how such rules are applied, categorized, and evaluated in practice.

Learning Objectives

After working through Section 5,

- You can **describe** what *flexible facility requirements* are and how they can be determined with a decision rule,
- you can **apply** a (simple) conditional-go decision rule to a demand scenario to determine a scenario-specific capacity vector; and
- you can **calculate** the net present value (NPV) and the expected net present value (ENPV) of a flexibly planned engineering system.

5.1 Flexible Facility Requirements

Facility requirements specify how, when, and, if relevant, where the capacity of an engineering system must be adjusted over time in order to meet future demand levels. As we already know from Section 4, for conventional strategic plans of engineering systems that are not designed with real options, facility requirements can be described by defining a capacity vector $\mathbf{K}$.

However, when engineering systems are designed with flexibility through real options, a fixed capacity deployment schedule $\mathbf{K}$ cannot be used to formulate facility requirements, since it is not known in advance when and how a real option included in the system will be exercised. For flexible engineering systems, an alternative approach for specifying facility requirements must therefore be adopted.

Decision Rules. According to the literature, one way to capture such *flexible facility requirements* is through the definition of a **decision rule** $\mathcal{D}$ (Cardin, 2014; Cardin & Hu, 2016; Cardin, Zhang, & Nuttall, 2017). At an abstract level, a decision rule is a mathematical function that maps a demand scenario $\mathbf{D}^s$ to an individual capacity vector $\mathbf{K}^s$ (Cardin & Hu, 2016):

$$\mathbf{K}^s = \mathcal{D}(\mathbf{D}^s, \boldsymbol{\theta}, \ldots) \tag{22}$$

Thereby, a decision rule emulates the decision-making of planners and owners who, rather than adhering rigidly to a plan established years in advance (as implied by conventional facility requirements), adapt to the situation at hand and make decisions based on the knowledge currently available (Cardin, Zhang, & Nuttall, 2017). In essence, a decision rule (i) prescribes the conditions under which a real option in an engineering system is to be exercised, and under which conditions it should remain dormant, and (ii) how the capacity of the engineering system is altered if the real option is exercised. Note: If an engineering system contains more than one real option, an individual decision rule must be formulated for each of them.

Decision rules map demand scenarios $\mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S$ to corresponding capacity vectors $\mathbf{K}^1, \mathbf{K}^2, \ldots, \mathbf{K}^S$ in an iterative manner. This means that, for a given demand scenario $\mathbf{D}^s$, the corresponding capacity vector $\mathbf{K}^s$ is obtained step by step: the system's capacity is first determined for planning stage $t = 1$, then based on the previous result capacity for planning stage $t = 2$ is determined, and so on, until reaching the planning horizon at $t = T$.

In each planning phase t, the capacity K_t^s of the engineering system under scenario $\mathbf{D}^s$ is defined as:

$$K_t^s = \mathcal{D}(\mathbf{D}_{[t]}^s, K_{t-1}^s, \boldsymbol{\theta}), \quad \forall t = 1, \ldots, T. \tag{23}$$

Equation 23 highlights that, for scenario s at time t, the decision on whether to exercise a real option depends on three pieces of information:

- The evolution of demand up to time t, expressed as

$$\mathbf{D}_{[t]}^s = [D_1^s, D_2^s, \ldots, D_t^s], \tag{24}$$

  which is often referred to as the *already disclosed uncertainty*;

- the capacity K_{t-1}^s already installed in the system in scenario s at time $t-1$;
- and, the parameterization $\boldsymbol{\theta}$ of the decision rule, specifying both the conditions for exercising the real option and the effect on the system's capacity when it is exercised.

Conditional-Go Decision Rules. Among the different types of decision rules mentioned in the literature (Cardin & Hu, 2016; Cardin, Xie, et al., 2017; Cardin, Zhang, & Nuttall, 2017; Zhang & Cardin, 2017) the *conditional-go* rule is the simplest. It takes the form of an **IF–THEN–ELSE statement**, commonly used in software programming, and specifies for a given real option: (i) the condition under which it is exercised (IF), (ii) the action to be taken if the condition holds (THEN), and (iii) the action if the condition does not hold (ELSE).

- **IF-condition:** tests whether the difference between installed capacity and observed demand is smaller than a threshold value θ_1 (which must be defined by the planners in advance):

$$\text{IF: } K_{t-1}^s - D_t^s < \theta_1 \tag{25}$$

- **THEN-statement:** states what happens if the IF-condition is true. In this case, the real option is exercised and the system's capacity increases, for example by value θ_2 (which depends on the nature of the real option):

$$\text{THEN: } K_t^s = K_{t-1}^s + \theta_2 \tag{26}$$

- **ELSE-statement:** states what happens if the IF-condition is false. In this case, the real option is not exercised and capacity remains unchanged:

$$\text{ELSE: } K_t^s = K_{t-1}^s \tag{27}$$

Illustrative Example. To better understand the functioning of conditional-go decision rules, let us consider the illustrative example shown in Figure 14. Assume we have a candidate flexibility for an engineering system consisting one single real option and a conditional-go decision rule $\mathcal{D}$ that specifies how the system's capacity changes depending on whether the option is exercised or not. Let us further assume that the option should only be exercised when capacity falls below demand, i.e., once the difference between K_{t-1}^s and D_t^s is negative (for this reason we set θ_1 in Equation 25 to 0). Besides, in case the option is exercised, the capacity of the system is changed by θ_2, see Equation 26.

To determine the capacity vector $\mathbf{K}^1$ with conditional-go decision rule $\mathcal{D}$ based on demand scenario 1, $\mathbf{D}^1$, we proceed as follows:

1. Test the IF-condition at $t = 1$. At $t = 1$, K^1_{t-1} corresponds to the system's initial capacity K_0. Because the difference between K_0 and D^1_1 is not negative (the pink dashed line in Figure 14 lies above the blue demand curve for $t = 1$), the IF-condition at $t = 1$ for scenario 1 is false. Consequently, the ELSE-statement applies, stating that capacity remains unchanged.

2. Increment to $t = 2$ and test the IF-condition again. If false, capacity remains unchanged.

3. Continue incrementing t until either the planning horizon T is reached or the IF condition becomes true. If true, we apply the THEN-statement, which dictates a capacity increase of:
$$K^s_t = K^s_{t-1} + \theta_2$$

4. Repeat this process iteratively until reaching the planning horizon $t = T$. The result of this process is a scenario-specific capacity vector $\mathbf{K}^s = \left[K^s_1, K^s_2, \ldots, K^s_T\right]$.

As illustrated in Figure 14, this iterative process is repeated for every demand scenario $\mathbf{D}^1, \ldots, \mathbf{D}^S$, so that each scenario is mapped to an individual capacity vector $\mathbf{K}^1, \ldots, \mathbf{K}^S$.

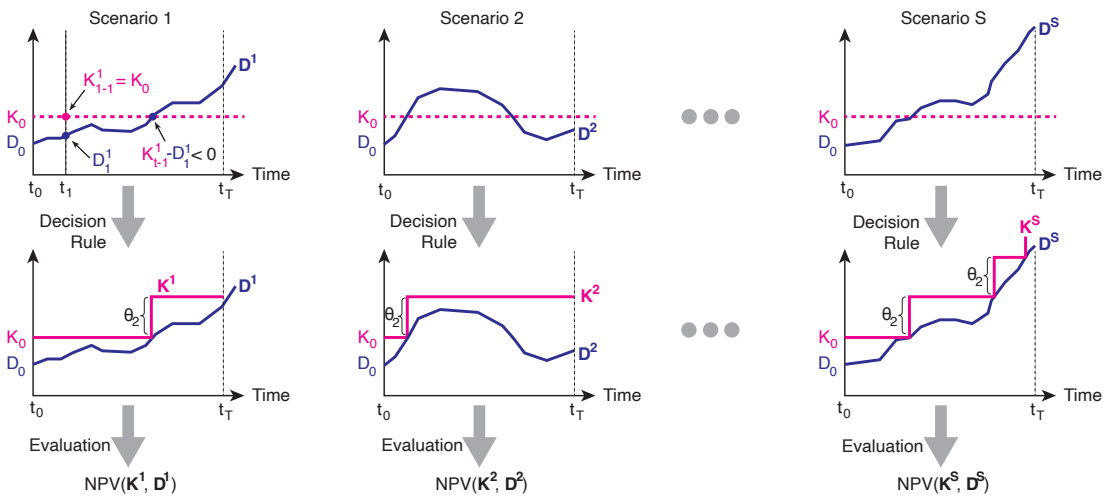

Figure 14: Illustration of a Conditional-Go Decision Rule Applied Iteratively Across Demand Scenarios

5.2 Evaluation of Flexible Facility Requirements

From Section 4, we know that a conventional facility requirement, expressed by a capacity vector $\mathbf{K}$, can be evaluated for a single demand scenario $\mathbf{D}^s$ in terms of its "quality" using the Net Present Value (NPV), see Equation 12. Likewise, across multiple demand scenarios $\mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S$, it can be evaluated using the Expected Net Present Value (ENPV), see Equation 21.

Equations 12 and 21 assume one single capacity vector $\mathbf{K}$. However, for flexible facility requirements in the form of a decision rule $\mathcal{D}$ this assumption does not hold anymore, as every demand scenario is mapped to an individual capacity vector $\mathbf{K}^1, \mathbf{K}^2, \ldots, \mathbf{K}^S$. Therefore, we have to find a way to evaluate the "quality" of flexible facility requirements.

As illustrated in Figure 14, the solution is rather straightforward: For each demand vector $\mathbf{D}^s$ and corresponding capacity vector $\mathbf{K}^s$, the resulting NPV of the system can be

determined by adapting Equation 12 as follows:

$$NPV^s(\mathbf{K}^s, \mathbf{D}^s) = \mathcal{C}_0 + \sum_{t=1}^{T} \frac{1}{(1+\delta)^t} \cdot \Big(\underbrace{\mathcal{R}_t(K_t^s, D_t^s) - \mathcal{C}_t(K_t^s, \Delta K_t^s, D_t^s)}_{K_t^s \text{ and } \Delta K_t^s \text{ result from decision rule application}} \Big). \quad (28)$$

Once the resulting $NPV^s(\mathbf{K}^s, \mathbf{D}^s)$ values have been calculated for all scenarios, the ENPV of the engineering system can be determined as follows:

$$ENPV(\mathbf{K}^1, \mathbf{K}^2, \ldots, \mathbf{K}^S, \mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S) = \sum_{s=1}^{S} p_s \cdot \underbrace{NPV(\mathbf{K}^s, \mathbf{D}^s)}_{\text{see Equation 28}}. \quad (29)$$

6 Selection of Optimal Facility Requirements

In Sections 4 and 5, we have seen how the "quality" of conventional and flexible facility requirements can be assessed using the Net Present Value (NPV) and the Expected Net Present Value (ENPV), respectively. We also know that the NPV and ENPV measure the value added to an engineering system when a conventional or flexible facility requirement is applied in planning: the larger the NPV or ENPV, the higher the "quality" of the facility requirement.

In this section, we address the question of whether it is possible to define an **optimal** facility requirement? This is precisely the focus of the so-called **capacity expansion problem** (CEP), which states that optimal facility requirements are those that maximize the Net Present Value of an engineering system when a single demand scenario is considered, or the Expected Net Present Value when multiple demand scenarios are taken into account (Luss, 1982; Martínez-Costa et al., 2014; Van Mieghem, 2003).

Learning Objectives

After working through Section 6,

- You can **explain** what optimal conventional facility requirements are and how they are defined;
- you can **explain** what optimal flexible facility requirements are and how they are defined;
- you can **describe** the conventional and the flexible capacity expansion problem; and
- you can **determine** "optimal" facility requirements (conventional and flexible) by **applying** the enumeration technique (for very simple cases).

6.1 Conventional Capacity Expansion Problem

Let us assume that we are developing a conventional strategic plan that takes into account a large number of different demand scenarios $\mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S$. In the conventional case, the CEP can then be formulated as the following optimization problem:

$$\max_{K} ENPV(\mathbf{K}, \mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S) \quad (30)$$

In simple terms, the optimization problem in Equation 30 states that we seek the conventional facility requirement (i.e., the capacity vector $\mathbf{K}$) that maximizes the ENPV of the system given the demand scenarios $\mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S$.

6.2 Flexible Capacity Expansion Problem

In the flexible case, the CEP is defined as follows:

$$\max_{\boldsymbol{\theta}} ENPV(\boldsymbol{\theta}, \mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S). \tag{31}$$

In this case, the parameterization $\boldsymbol{\theta} = [\theta_1, \theta_2]$ of the conditional-go decision rule is sought, which maximizes the ENPV of the system given the demand scenarios $\mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S$. In other words, the optimization problem consists of determining (i) the *optimal* IF-condition (i.e., θ_1), and (ii) the optimal "size" of the real option (i.e., the optimal capacity increase θ_2).

6.3 How to Solve the CEP?

While the conventional CEP and the flexible CEP are conceptually straightforward, solving them is considerably more challenging. The literature presents various, and in some cases highly complex, solution methods for identifying the optimal capacity vector $\mathbf{K}^*$ or the optimal parameterization of the decision rule $\boldsymbol{\theta}^*$[3] These solution approaches go far beyond the scope of this course. For this reason, we will focus only on a single, very simple and admittedly not very elegant solution method, which is commonly known as the **enumeration technique**.

The enumeration technique is based on the *brute force* concept. In a first step, all potentially possible **solution candidates** must be determined. In the case of the conventional CEP, this means enumerating all theoretically possible capacity vectors, while in the case of the flexible CEP it requires enumerating all theoretically possible parameter vectors.

To make the enumeration technique more concrete, let us consider a simple example of a conventional CEP. Assume we want to solve a conventional CEP for a relatively short planning horizon, resulting in only two planning phases, t_1 and t_2. Starting from an initial capacity of $K_0 = 1$, we assume that the system's capacity can only be changed from one planning stage to the next by either $\Delta K = 0$ or $\Delta K = 2$. Thus, there exist only four possible capacity vectors:

$$\mathbf{K}^1 = [1, 1]\,, \quad \mathbf{K}^2 = [1, 3]\,, \quad \mathbf{K}^3 = [3, 3]\,, \quad \mathbf{K}^4 = [3, 5]\,.$$

The first solution candidate, $\mathbf{K}^1$, assumes that capacity remains unchanged in both planning phases. $\mathbf{K}^2$ assumes that capacity is increased by $\Delta K = 2$ in the second planning phase, while $\mathbf{K}^3$ assumes an increase of $\Delta K = 2$ in the first planning phase only. Finally, solution candidate $\mathbf{K}^4$ assumes that capacity is increased by $\Delta K = 2$ in both planning phases.

Once we have *enumerated* all possible solution candidates, the second step is to compute the resulting ENPV of the system for each candidate. Suppose we obtain the following results:

[3] An asterisk denotes the *optimal solution* of the corresponding optimization problem.

$$
\begin{aligned}
\text{ENPV}_1(\mathbf{K}^1, \mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S) &= 369.7, \\
\text{ENPV}_2(\mathbf{K}^2, \mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S) &= 115.7, \\
\text{ENPV}_3(\mathbf{K}^3, \mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S) &= 281.7, \\
\text{ENPV}_4(\mathbf{K}^4, \mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S) &= -105.4
\end{aligned}
$$

In the third and final step, we select the optimal solution candidate, which is the solution candidate that yields the highest ENPV. In this example, the optimal solution is $\mathbf{K}^* = \mathbf{K}^1$.

As this example illustrates, the enumeration technique has one major drawback: all possible solution candidates must be explicitly generated and tested. The enumeration method is therefore only computationally feasible when the number of potential solutions candidates is reasonably small and manageable.

6.4 Implications of the CEP

The CEP enables the definition of optimal facility requirements, and thus optimal strategic plans, for both conventional and flexible cases. These plans are considered optimal because they maximize the value of the engineering system across the set of future demand scenarios $\mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^S$. In this way, the CEP provides planners and owners of engineering systems with a tool to design strategic plans based on objective and measurable criteria.

A particularly important implication arises from the role of optimal flexible facility requirements in the strategic planning process. While a conventional strategic plan can be likened to a timetable that specifies precisely when and how system capacity should be adjusted, flexible facility requirements instead define optimal decision rules. This represents a paradigm shift in strategic planning: the focus is no longer on prescribing an optimal schedule, but on identifying what constitutes optimal decisions under uncertainty.

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